

CURRENT FINAL TERM

BT503

EFFORT BY: WARRIORS + ASHAM

1. Consequences of OECD 2001 about global environment 3

The OECD (2001) concluded that the industrial use of biotechnology commonly leads to increase harmonious processes and results in lowered operating and capital costs.

2. What are qualities of pollutant that are intended target in bio processing? 5

The intended target of the bio-processing must be both susceptible and available to biological attack, in aqueous solution, or at least in contact with water, and within low to medium toxicity range. For land applications, the soil types best suited to biotechnological interventions are sands and gravels. Moreover, nutrient availability, oxygenation and the presence of other contaminants play a role too.

3. Extremophile 6

In general use of biotechnology for environmental that relies on mesophilic microorganisms. Some microbial species tolerate extreme conditions, like high salinity, pressures and temperatures, useful for biotech applications that tolerance to these conditions

The Archaea

They are extreme thermophiles, extreme halophiles, and methanogens, species that tolerate high levels of radiation, pH, or high pressure. Extremophiles provide way of developing alternative routes to many conventional chemicals or material

An extremozyme is an enzyme that created by archaea, which are known prokaryotic extremophiles that can function under extreme environments where other enzymes not survive.

4. Thermopolis found 7

A thermophile is an organism, type of extremophile that thrives at temperatures above 45°C. Many thermophiles are archaea, though they can be bacteria. Hot-springs, deep sea-vents, geothermal fluids, etc.

5. Acidophile 8

Acidophiles thrive in the conditions of low pH. Protective molecules present in their cell walls, membranes or outer cell coatings to exclude acids

6. Endocrine disrupter 10

There are chemicals which resist degradation. Partly due to lack of organisms with degradative abilities, or worse due to microbial activities that change them to more toxic form

7. Difference between transductions, transformation, plasmid and conjugation. 12

Transformation: In transformation, a bacterium takes piece of DNA floating in its environment.

Transduction: In transduction, DNA is accidentally moved from one bacterium to another by a virus.

Conjugation: In conjugation, DNA is transferred between bacteria through a tube between cells.

Plasmid: A plasmid is a small, extra chromosomal DNA molecule within a cell that is physically separated from chromosomal DNA and can replicate independently.

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8. Catabolic expansion? 14

Improving microbe's ability to degrade contaminant by culturing the bacteria in growth medium in which the contaminant is supplied as an essential part of the nutrition. Only bacteria which have undergone a mutation enabling them to utilize this food source will be able to survive.

9. Fundamental requirements of all cloning techniques 16

- ♥ The enzymes, solutions and equipment necessary to perform the procedures
- ♥ The desired piece of DNA to be transferred
- ♥ a cloning vector
- ♥ The recipient cell.

10.cDNA libraries 18

In eukaryotes, the first product of transcription from DNA is not messenger RNA but heterogeneous nuclear RNA (hnRNA). This is mRNA prior to the removal of all the non-coding sections or introns which are discarded during the processing to produce the mature mRNA. cDNA is DNA which has been artificially made by mature mRNA as a template

11.Blotting northern, Western 21

Northern blotting detects a specific RNA sequence from an RNA sample.

Southern blotting detects a specific DNA sequence from a DNA sample.

Western blotting detects a specific protein from a protein sample by using Agarose /formaldehyde gel

12.Transgenic plant 24

“A transgenic plant is a modified organism where genes are transferred from one organism to another through genetic engineering techniques”.

13.Ways to improve crop quality 24

Crop improvement refers to the ways of developing and breeding of crop varieties which are better than the pre-existing cultivars in a number of characters like taste, height, colour, etc. for a better yield. Many methods used to improve crop performance. Others involve improvements in farm practices, irrigation, and drainage, and herbicide, pesticide and fertiliser use.

14.T DNA 26

The transfer DNA is the transferred DNA of the tumor-inducing (Ti) plasmid of bacteria such as *Agrobacterium tumefaciens* and *Agrobacterium rhizogenes*. The T-DNA is transferred from bacterium to host plant's nuclear DNA genome.

15.How plants protect themselves from TMV 28

For protection of plant introduction of the genes expressing antibodies to protein of Tobacco Mosaic Virus (TMV) by *A.tumefaciens*. Expression of genes in the plant leads to complete immunity against TMV.

16.Quorum sensing 29

Bacteria communicate with each other by way of small diffusible molecules such as the Nacylhomoserine lactones (AHLs) is called quorum sensing.

17.Natural processes to remove the pathogens organism. 29

Natural: Maturation Ponds, Land treatment

Maturation ponds: penetration of UV radiation and unfavorable environmental conditions causes a high mortality of the pathogens.

Land treatment: The unfavorable environmental conditions in the soil favor the mortality of the pathogens. Chemical products are not needed. Requires large areas

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18.Maturation pond

The main objective is to removal pathogenic of organisms. Environmental conditions, such as UV radiation, high pH, high DO, lower temperature, lack of nutrients, etc. Maturation ponds are a post-treatment stage for BOD-removal processes. The maturation pond is designed in such a way to optimize various factors that contribute to the death of the pathogens such as temperature, solar radiation, pH, food shortage, predator organisms, competition, toxic compounds, etc. Many of these mechanisms become more effective with lower pond depth.

19.What are cynophyta? 34

Cyanophyta is also referred to as Cyanobacteria that are a phylum of bacteria that obtain their energy through photosynthesis. It is type of phytoplankton

20.Bio indicators 33

Bioindicators are living organisms such as plants, planktons, animals, and microbes, which are utilized to screen the health of the natural ecosystem in the environment.

21.How plants are modified as bioindicator 35

The presence or absence of some plants or other vegetation provides information about environmental health. e.g., Lichens generally found on the trunks of trees and rocks, react to ecological changes in forests, including changes in the structure of the forest, air quality, and climate

22.Phytoplankton 41

Phytoplanktons, also known as microalgae, are similar to terrestrial plants in that they contain chlorophyll and require daylight to live and develop. They swim in the upper portion of the sea, where light infiltrates the water

23.Zoo planktons 43

The word "zooplankton" is derived from the Greek zoon, meaning "animal", and planktos, meaning "wanderer" or "drifter".

Zooplanktons are microscopic animals living near to the surface of the water body. They are poor swimmers, relying on tides and currents as a transport mechanism. They feed upon phytoplankton, bacterio-planktons, or detritus. They also play an important role as Bioindicators and help to determine the level of water pollution.

24.Environment pollution? 46

Any change in our environment that has harmful effects on plants, animals and human is called environmental pollution.

25.Role of filter medium in boiler also name the compound of the filter medium. 60

The first methods to be developed that consists of large vessel or container. They made of cast concrete, metal or plastic, which holds a filter medium of organic material such as peat, heather, bark chips.

The medium offers good water holding capacity and soluble chemicals within the waste gas dissolve into the film around the matrix. Bacteria, and other micro-organisms present that degrade components of solution and bring the desired effect. The medium provides physical support for microbial growth, with a large surface area, high in internal empty spaces and rich in nutrients to stimulate bacterial activity. Bio filters need much watered to maintain internal conditions, but waterlogging is to be avoided due to compaction, and reduced efficiency. Properly maintained, bio filters can reduce odor release by 95% or more.

26.Biofilters and its phases or phases of aerated biofilter 60

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Biofiltration is a pollution control technique using a bioreactor containing living material to capture and biologically degrade pollutants.

This bio-filter is a three-phase reactor composed of:

- ♥ Solid phase: consists of a support medium and biofilms
- ♥ Liquid phase: consists of the liquid in permanent flow through the porous medium
- ♥ Gas phase: formed by artificial aeration and by the gaseous by-products of the biological activity.

27.Five advantages of bio filter 60

- ♥ Low investment
- ♥ Simple construction
- ♥ Effective removal of biologically degradable components
- ♥ Low pressure drop
- ♥ Little waste water
- ♥ Little waste material

28.Bio scrubber 62

The bio scrubber is not itself a biological treatment system, but highly efficient method of removing odour components by dissolving them. Most appropriate for hydrophilic compounds like acetone or methanol.

29.Advantages of trickling filter over biofilter 62

The efficiency of the bio trickling filter is broadly similar to the bio filter it can deal with higher concentrations of contaminant and has smaller foot-print than a biofilter. Although trickling filters are more easily operated and consume less energy than activated sludge processes, they have lower removal efficiency for solids and organic matter, they are more sensitive to low air temperatures

30.Difference between adsorption and incineration 64

Absorption is a chemical or physical process in which the molecules, atoms and ions of the substance getting absorbed enter into material in which it is taken up.

Incineration refers to the destruction of material such as waste. Incineration is the high-temperature burning of a waste

31.Biomimetic 78

“Biomimetics is an interdisciplinary field in which principles from engineering, chemistry and biology is applied to the synthesis of materials, systems or machines that have functions that mimic biological processes.”

32.Waste water treatment 81

Wastewater treatment is the process of converting wastewater to water that is no longer needed or no longer suitable for use that can be discharged back into the environment. It's formed by a number of activities like bathing, washing, using the toilet, and rainwater runoff.

33.Advantages of conventional biological sludge 111

The biological stage comprises two units: aeration tank (reactor) and sedimentation tank.

- ♥ Biomass remains in the system longer than the liquid
- ♥ Resulting in high BOD removal efficiency
- ♥ Low cost
- ♥ Good quality effluent
- ♥ Small land area require

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- ♥ Loss of head is small
- ♥ Freedom from fly and odor
- ♥ Easily maintained mechanical work
- ♥ Self-sustaining system

34.Biodisc 119

Bio-disc technology is a wastewater treatment process involving aerobic biological digestion with fixed biomass.

35.Main purpose of bio disc 119

- ♥ to serve as the surface for microbial film growth
- ♥ to promote the contact between the microbial film and the sewage
- ♥ To maintain the biomass that detached from the discs in the liquid
- ♥ To promote the aeration of the sewage that is agree to the disc

36.Stages of Sludge waste management 128

The main stages in sludge management:

- ♥ Thickening: removal of water (volume reduction)
- ♥ Stabilization: removal of organic matter (mass reduction)
- ♥ Conditioning: preparation for dewatering
- ♥ Dewatering: removal of water (volume reduction)
- ♥ Disinfection: removal of pathogenic organisms
- ♥ Final disposal: final destination

37.Difference between physical and mechanical sludge dewatering process 128

Sludge is separated in a solid and a liquid part with the help of a physical process such as filtration, squeezing, centrifugal separation or compaction.

Mechanical dewatering is mostly applied for the treatment of sludge in large-scale wastewater treatment plant.

38.Sludge dewatering 130

Sludge dewatering is practice of minimizing waste by volume to prepare for effective disposal.

39.dewatering main objective 130

The main objectives:

- ♥ Reduction of the transportation costs
- ♥ Improvement of the handling conditions
- ♥ Increase of the calorific value of the sludge (for incineration)
- ♥ Reduction in the volume for disposal
- ♥ Reduction in the production of leachate when the sludge is disposed of in landfills

40.natural and mechanical dewatering difference 130

Sludge dewatering can be done by natural or mechanized processes.

Natural processes use evaporation and percolation as the main water removal mechanisms, thus require more time for dewatering. Although simpler and cheaper to operate, they need larger areas and volumes for installation.

Mechanized processes are based on mechanisms such as filtration, compaction, or centrifugation to accelerate dewatering. In terms of operation and maintenance, this leads to small and complex equipment.

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41.Mention the most common metallic coagulants for sludge dewatering 130

Aluminium sulphate • ferric chloride • ferrous sulphate • ferric sulphate • quicklime/ hydrated lime

42.Advantage and disadvantage of incineration 135

Advantages: • Drastic volume reduction • Sterilization

Disadvantages: • High costs • Ash disposal • Atmospheric pollution

43.Disadvantage of in situ and ex situ? 138

Ex situ Disadvantages: Additional transport costs, * increased likelihood of spillage

In situ disadvantages: Stringent requirement for site investigation and survey. Because it is difficult to regulate the reaction conditions, the 'optimisation' of the process may not actually be optimal.

44.Characteristics of intensive technology 139

Intensive technologies can be characterized as fast acting, high intervention strategies, with a heavy demand for resources and high initiation, running and support costs. Their key factors are a fast response and low treatment time, which makes them excellent for heavy contamination conditions. E.g., Soil washing and thermal treatments

45.Bioremediation and its types 141

Bioremediation is the use of microbes to clean up contaminated soil and groundwater. Microbes are very small organisms, such as bacteria, that live naturally in the environment.

Three major routes of Bioremediation:

- ♥ Mineralization
- ♥ Co-metabolism
- ♥ Immobilization.

46.Factors effect bioremediation 142

Environmental Factors:

Temperature: 0 and 50 °C. Most efficient: 20–30 °C

PH: pH range of 5.0-9.0, however, 6.0-8.0 being most efficient.

Soil type: sands and gravels are the most suitable soil types for bioremediation

Other factors:

Nutrient availability, oxygenation, the groundwater runs off, other contaminants that are present, concentration of contaminants and place where they are biodegradable, etc.

47.Bio augmentation and bio enhancement 143

Bioenhancement concentrates solely on micro-faun that stimulating activity by the manipulation of local environmental conditions.

Bioaugmentation: “the introduction of cultured microorganisms into environment in order to enhance bioremediation of pollutants.

48.Phyto-remediation 150

“Direct in situ use of living green plants for the treatment of soil, sludges or ground water that remove degradation or containment of the pollutants present”

49.Photovoltlization 153

It is involves the uptake of the contaminants by plants and release into the atmosphere in a modified form. Fast-growing trees relies on the transpiration pull which accelerates the uptake pollutants in ground water solution then transformed and released through the leaves

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E.g. Genetically modified variety of the Yellow Poplar, *Liriodendron tulipifera* engineered by the introduction of mercuric reductase gene. This is the ability to tolerate higher mercury concentrations, and removal of the pollutant from the soil and volatilize it.

50. Phytodegradation and its forms 153

Phytodegradation involves the degradation of organic contaminants directly, through the release of enzymes from roots, or through metabolic activities within plant tissues.

Forms are:

- ♥ Internally, take up by the plants,
- ♥ Externally, using enzymes secreted by them

51. Enlist sources of solid waste 162

- ♥ Organic waste: Waste from preparation of food, market places, etc.
- ♥ Combustibles: Paper, wood, etc.

Different categories of solid waste include:

- ♥ Non-combustibles: Metal, bottles, stones, etc.
- ♥ Ashes/dust: Residue from fires used for cooking.
- ♥ Bulky waste: Tree branches, tyres, etc.

Different categories of solid waste include:

- ♥ Dead animals: dead body of animals and livestock.
- ♥ Hazardous waste: Oil, battery acid, medical waste.
- ♥ Construction waste: rubble, broken concrete, etc.

52. Fermentation produce solution of ethanol in water list two methods 176

Ethanol is produced by fermentation and concentrated by distillation. Fuel-grade ethanol produced by simple distillation, to 95% ethanol, or to the anhydrous form by azeotropic co-distillation using a solvent

53. Advantages of biodiesel 179

Biodiesel is derived mostly from vegetable oils

Trans-esterification

Biodiesel use as vehicle fuel increases energy security, improves air quality and the environment, and provides safety benefits. Biodiesel exhaust is generally less harmful to both human health and the planet.

54. Advantages of biofuels 180

Enormous energy demand in industrialized countries

Biofuels help to reduce the carbon footprint of transportation

Every gallon of biofuel replaces a gallon of fossil fuel helps reduce greenhouse-gas emissions.

55. Bio pesticides? 183

Biopesticides are certain types of pesticides derived from natural materials as animals, plants, bacteria, and certain minerals. They use for manage plant pathogens, invertebrates pests etc.

56. Association of fungi with plants 186

Mycorrhizae: Associations of fungi with roots of vascular plants common and in some cases beneficial to the plant. They may be external, ectomycorrhizal, or internal, endomycorrhizal.

57. Minimum substrate concentration 200

Biodegradation of organic compounds is affected by S_{min} . Lower concentrations with slower reaction cannot provide energy to organism for growth. Many dangerous organic compounds

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drink water at low rate. Such concentrations are less than 5 mg/L while compound may be easily biodegradable at higher concentrations.

58. Main chemicals characteristics of chlorinated pesticides that lead to problems 204

Two characteristics of chlorinated pesticides that led to great harm are their resistance to biodegradation and hydrophobicity.

59. Three Sources of dioxins 220

Dioxins are mainly byproducts of industrial practices. They are produced through a variety of incineration processes, improper municipal waste incineration and burning of trash, and can be released into the air during natural processes, such as forest fires and volcanoes.

60. dioxin effects on human 220

Human responses include weight loss, insomnia, liver dysfunction, spontaneous abortion, and cancer.

61. Biosensor and its advantages 228

A biosensor is an analytical device, used for the detection of a chemical substance that combines a biological component with a physicochemical detector

Advantages:

Biosensors monitor the presence of various contaminants in order to ensure the quality of drinking water, food, and soil.

Biosensor is used to break down the glucose level in the blood of an individual.

62. With the help of named example discuss how whole organism can be used to measure potential biological toxicity of water and soil 230

Much effort use to develop and use different bio assays and biosensors for toxicity of water samples. Whole organisms to measure the potential biological toxicity of a water or soil sample. Microtox® or ToxAlert® toxicity assays use the luminous *Vibrio fischeri* bacteria to detect the toxicity of substances. For quick investigation, use Cellsense®, that contains *Escherichia coli* cells.

63. What are pcbs? which biosensors detect them 234

Polychlorinated biphenyls (PCBs) are universal environmental pollutants. They are widely used as industrial chemicals, as dielectric fluids in electrical transformers and capacitors. The production of PCBs has been banned but these are still present in the environment.

Different biosensor determines PCBs in the environment. DNA biosensor with chronopotentiometric detection, various immune-sensors with fluorescence, SPR, and electrochemical detection principles use for this determination.

64. Benefits of fungi to plant

Algae possess the ability to extract nitrogen from the air and deposit the nitrogen into the surrounding soil or water. In addition to improving plant growth, algae suppress soil-borne plant pathogens and repel pests, which decrease the need of chemical fertilizers and dangerous pesticides.

65. Characteristics of optical transducer

Optical transducer converts light rays into an electronic signal. The sensor detects wavelength, frequency, or polarization of light and converts it into electric signal due to photoelectric effect.